

Level Sets

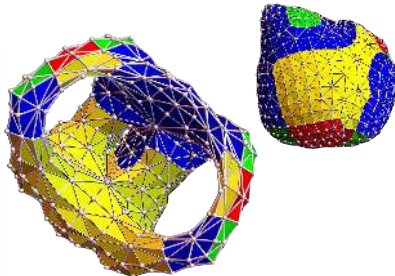
G: finite abstract simplicial complex with vertex set V and dimension q.

$$g : V \rightarrow K_k = \{0, \dots, k\} \text{ function}$$

$$M_g = \{x \in G, g(x) = K_k\}$$

M_g is open but defines a graph that has a simplicial complex as Whitney complex.

For any q-manifold G and any function $g : V \rightarrow K_k$, the level set M_g is a (q-k)-manifold or empty.



Proof

Induction in q. Use hyperbolicity:

$$S(x) = S^+(x) \oplus S^-(x)$$

for a n-simplex x

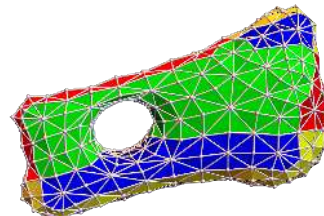
$S^-(x)$ is a (n-1)-sphere, the boundary complex of x.

A $S_g^-(x)$ is a (n-k-1)-sphere

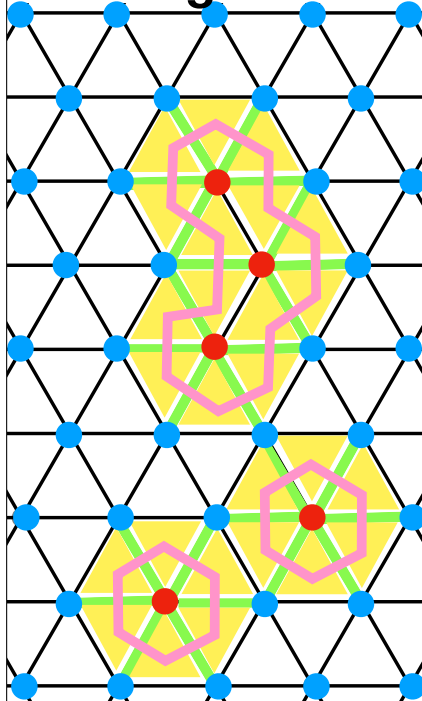
B $S_g^+(x) = S^+(x)$ (q-n-1)-sphere

Proof: $S_g^+(x) \subset S^+(x)$
 $y \in S^+(x)$ implies $y \in S_g^+(x)$

C $S_g(x) = S_g^+(x) \oplus S_g^-(x)$
 is a (q-k-1)-sphere

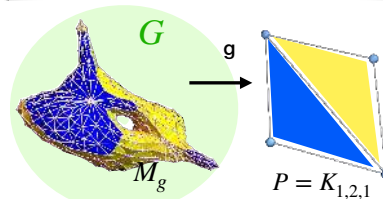


No singularities!



Generalization

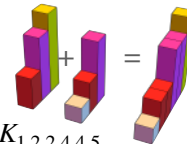
G: q-manifold with vertex set V.
 $g : V \rightarrow P = K_{n_0, \dots, n_k}$ partition complex
 $M_g = \{x \in G, g(x) \text{ contains facet in } P\}$



If not empty, M_g is a (q-k)-manifold. It is union of patches, belonging to facets reached by g.

Partition complexes form a commutative join monoid

Example:

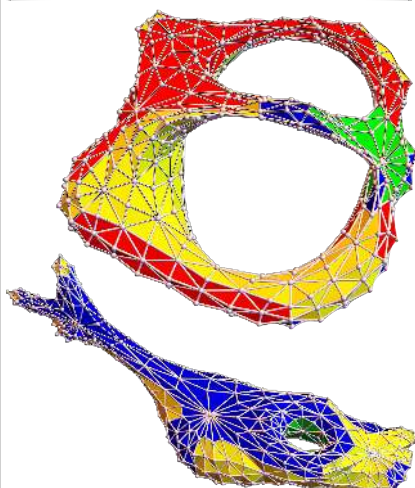


$$K_{2,4,5} \oplus K_{1,2,4} = K_{1,2,2,4,4,5}$$

Code

The implementation is straightforward:

```
Generate[A_] := If[A == 0, 0, Sort[Delete[Union[Sort[Flatten[Map[Subsets, A], 1]], 1]], 1];
Whitney[s_] := Generate[Sort[Flatten[Map[Subsets, A], 1]], 1];
RFunction[G, f_] := Module[{Rf, l := 0}, RandomChoice[Range[1, Ver[G]], Map[Rf, Ver[G]]];
Facets[G_] := Select[G, {l, l} == Map[Map[G], G] &]; PolyX[_] := PolyhedronData[X, "Skeleton"];
AbstractSurface[G, f, A_] := Select[G, {Sum[{Subsets[G], f, A], 1, 0}, {l, l} > 0} &];
Zf[_] := Partition[Range[1, 1], ZeroJoin[k_] := If[k] == 1, 2, 0]; Whitney[CompleteGraph[n]]];
CJoin[G, H, l_] := Union[G, H + Map[G], 1, Map[Flatten, Map[Union, Flatten[Tuples[G, H + Map[G], 1], 0]], 1];
ToGraph[G_] := UndirectedGraph[n := Length[G], Graph[Range[n], Select[Flatten[Tuples[G, H + Map[G], 1], 0]], 1];
Subsets[G][M][2][3], G][k] &];
l := Whitney[Poly[ToGraph[G]]; G][CJoin[l, Whitney[CycleGraph[13]], PzZeroJoin[2, 2, 3]];
V := Ver[P]; F := Facets[P]; H := AbstractSurface[G, RFunction[G, V], F]; GraphPlot3D[ToGraph[H]]
```



Random manifolds

